Decent Coaching Classes

VIII SB

Mind Your Math - IX

ANSWER KEY & SOLUTIONS

Q.1 (A) MCQ (1 mark each) **[8]**

(1) C (2) A (3) B (4) C (5) B (6) B (7) D (8) C

Hints: (2) $-2/3 = -8/12$, $-3/4 = -9/12$; for negatives the number with the smaller positive value is greater, so $-2/3 > -3/4$.

(3) Interior angles are supplementary: $180° - 75° = 105°$. (4) Corresponding angles are equal: $2x + 10 = 3x - 20$, so $x = 30$.

(5) $0.000343 = 343/10^6$; cube root $= 7/100 = 0.07$. (6) $32^{3/5} = (32^{1/5})^3 = 2^3 = 8$.

(8) The centroid divides the median in the ratio 2 : 1, so $l(AG) = (2/3) \times 18 = 12$ cm.

Q.1 (B) Fill in the blanks / True or False (1 mark each) **[8]**

(1) non-recurring (2) transversal (3) $243^{1/5}$ (4) right (5) exterior

(6) True (7) False (cube root of -27 is -3) (8) False ($x \times y$ is constant, not $x \div y$)

Q.2 Any FIVE (2 marks each) **[10]**

(1) $(2p + 5)(2p - 3) = (2p)^2 + (5 - 3)(2p) + (5)(-3) =$ **$4p^2 + 4p - 15$**

(2) $(y + 5)^3 = y^3 + 3(y^2)(5) + 3(y)(5)^2 + 5^3 =$ **$y^3 + 15y^2 + 75y + 125$**

(3) Factors of 28 with sum 11 are 4 and 7. $x^2 + 11x + 28 = x^2 + 4x + 7x + 28 = x(x + 4) + 7(x + 4) =$ **$(x + 4)(x + 7)$**

(4) $x^3 - 125 = x^3 - 5^3 =$ **$(x - 5)(x^2 + 5x + 25)$**

(5) $m \propto n$, so $m = kn$. $12 = k \times 4$, so $k = 3$. Constant of variation = **3**; equation of variation: **$m = 3n$**

(6) Diagonals of a rhombus bisect each other at 90°. Half-diagonals = 12 cm and 5 cm. Side$^2 = 12^2 + 5^2 = 144 + 25 = 169$, so side = **13 cm**

(7) Discount = 12% of 1500 = 180. Selling price = $1500 - 180 =$ **₹ 1320**

Q.3 Any FOUR (3 marks each) **[12]**

(1) $(3x - 2y)^3 = (3x)^3 - 3(3x)^2(2y) + 3(3x)(2y)^2 - (2y)^3 =$ **$27x^3 - 54x^2y + 36xy^2 - 8y^3$**

(2) $(2p + q + 3)^2 = (2p)^2 + q^2 + 3^2 + 2(2p)(q) + 2(q)(3) + 2(2p)(3) =$ **$4p^2 + q^2 + 9 + 4pq + 6q + 12p$**

(3) $2 \times (-15) = -30$; factors with sum 7 are 10 and -3. $2x^2 + 10x - 3x - 15 = 2x(x + 5) - 3(x + 5) =$ **$(x + 5)(2x - 3)$**

(4) $54a^3 - 2 = 2(27a^3 - 1) = 2[(3a)^3 - 1^3] =$ **$2(3a - 1)(9a^2 + 3a + 1)$**

(5) Workers and days are in inverse variation, so workers × days = k. $k = 12 \times 20 = 240$. For 15 days: $n \times 15 = 240$, so $n = 16$. **16 workers** are needed.

(6) Selling price $= 25 \times 2400 =$ ₹ 60,000. Commission = 3% of 60,000 = **₹ 1800**. Amount received = $60,000 - 1800 =$ **₹ 58,200**

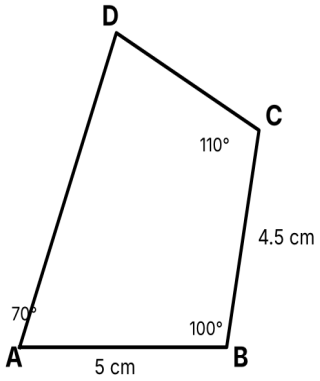
Q.4 Any THREE (4 marks each) **[12]**

(1) $(m + 3)^3 = m^3 + 9m^2 + 27m + 27$; $(m - 3)^3 = m^3 - 9m^2 + 27m - 27$. Subtracting: **$18m^2 + 54$**

(2) $x^2 - 5x + 6 = (x - 2)(x - 3)$; $x^2 - 9 = (x - 3)(x + 3)$. So the expression = **$(x - 2)/(x + 3)$**, $x \neq 3, -3$

(3) $p \propto 1/q^2$, so $p \times q^2 = k$. $k = 5 \times 16 = 80$. (i) $q = 2$: $p \times 4 = 80$, so **$p = 20$**. (ii) $p = 1.25$: $1.25 \times q^2 = 80$, so $q^2 = 64$, **$q = 8$** (or -8).

(4) Rough figure and steps: (1) Draw seg AB = 5 cm. (2) At A draw a ray making 70° with AB. (3) At B draw seg BC = 4.5 cm making 100° with BA. (4) At C draw a ray making 110° with CB, on the same side as the ray from A. (5) The two rays meet at D. □ABCD is the required quadrilateral. (∠D = 80°; AD ≈ 6.8 cm, CD ≈ 4.0 cm.)



Marking: 1 mark rough figure, 1 mark each for correct arcs/angles at A and B with AB, BC, 1 mark for the angle at C and the final quadrilateral.

(5) Let the decided price be ₹ 100. Marked price = 100 + 30 = ₹ 130. Discount = 10% of 130 = ₹ 13. Selling price = 130 − 13 = ₹ 117. Gain = 117 − 100 = 17, so the percentage gain on the decided price is **17%**.

Q.5 Any TWO (5 marks each) [10]

(1) Adjacent angles of a parallelogram are supplementary. (i) (3x + 10) + (2x + 20) = 180, so 5x = 150, **x = 30**. [2]

(ii) ∠P = 3(30) + 10 = 100°; ∠Q = 2(30) + 20 = 80°. Opposite angles are equal, so ∠R = 100° and ∠S = 80°. Angles: **100°, 80°, 100°, 80°** [2]

(iii) Diagonals of a parallelogram bisect each other. l(QT) = ½ × l(QS) = ½ × 14 = **7 cm** [1]

(2) Discount = 8% of 2500 = ₹ 200. Selling price = 2500 − 200 = ₹ 2300. [2]

Profit is 15%: when cost price is ₹ 100, selling price is ₹ 115. Let the cost price be ₹ x for selling price ₹ 2300: x/100 = 2300/115, so x = 2300 × 100 / 115 = **₹ 2000** [3]

(3) (i) $(102)^3 = (100 + 2)^3 = 100^3 + 3(100)^2(2) + 3(100)(2)^2 + 2^3 = 1{,}000{,}000 + 60{,}000 + 1200 + 8 =$ **1,061,208** [2]

(ii) 6 × (−15) = −90; factors with sum 1 are 10 and −9. $6x^2 + 10x - 9x - 15 = 2x(3x + 5) - 3(3x + 5)$ = **(3x + 5)(2x − 3)** [3]

Note: Award full marks for any correct alternative method.